Permutations

The number of ways to arrange r objects out of n is where **order matters** is...
The number of r length words over an alphabet with n letters **without** repetitions

$$\begin{aligned} n &: \text{Total \# of objects} \\ r &: \text{\# of objects you want to choose from} \end{aligned}$$

$$\begin{aligned} P(n,r) = {}_nP_r &= n(n-1)(n-2)\cdot\ldots\cdot(n-r+1) \\ &= \frac{n!}{(n-r)!} \end{aligned}$$

Some examples are:

- Different ways of choosing r roles out of a class of n students, where each role has one student playing it, and no students have multiple roles
- Different ways of passing out 5 cards to a group of 5 people out of a deck of 52, where each person gets one card
- Different ways of having 1st 2nd 3rd place runners in a race of 10 athletes.
- Different ways to make a 4-letter word using different letters.

Combinations

1. How many ways can I pick r objects from n distinct objects in an **unordered** set?
2. r-element subsets of a set of cardinality n
3. Length n binary strings of length r

$$C(n,r) = {}_nC_r = \binom{n}{r} = \frac{n!}{r!\cdot(n-r)!}$$

Stars and Bars

$$\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$$

$n:$ number of items
$k:$ number of bins

Where

1. Order doesn't matter
2. Bins can contain 0
3. All items must be placed

Integer Equations

$$\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$$

$n:$ RHS
$k:$ Number of variables

$$\binom{n+k(1-m)-1}{k-1}$$

$n:$ Number of items
$k:$ Number of bins
$m:$ At Least

For

$$(a_1+c_1)+(a_2+c_2)+\ldots(a_k+c_k)=n$$

This is just saying that $a_i \geq c_i$, so just move all the constants to the left and solve like normal

$$a_1+a_2+\ldots+a_k = n-c_1-c_2-\ldots-c_k$$

Binomial Theorem

$$\text{For } n=0,1,2,\ldots,(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$$

Inclusion Exclusion (PIE)

Let $A_1, A_2, \ldots, A_n$ be finite sets. Then

$$\begin{aligned} |A_1\cup A_2\cup\ldots\cup A_n| &= \sum_{1\leq i\leq n}|A_i| \\ &- \sum_{1\leq i<j\leq n}|A_i\cap A_j| \\ &+ \sum_{1\leq i<j<k\leq n}|A_i\cap A_j\cap A_k| \\ &-\ldots+(-1)^{n+1}|A_1\cap A_2\cap\ldots\cap A_n| \end{aligned}$$

ODD sets involved: **ADD**
EVEN sets involved: **SUBTRACT**

Cardinality of Intersections

Let $A_1, A_2, \ldots, A_n$ be finite sets. Then

$$\begin{aligned} |A_1\cap A_2\cap\ldots\cap A_n| &= |U| - \sum_{i=1}^{n}|\overline{A_i}| \\ &+ \sum_{k_1<k_2}|\overline{A_{k_1}}\cap\overline{A_{k_2}}| \\ &- \sum_{k_1<k_2<k_3}|\overline{A_{k_1}}\cap\overline{A_{k_2}}\cap\overline{A_{k_3}}| \\ &+\ldots+(-1)^n|\overline{A_{k_1}}\cap\overline{A_{k_2}}\cap\ldots\cap\overline{A_n}| \end{aligned}$$

The big idea is

ODD number of sets, then **SUBTRACT**
EVEN number of sets, then **ADD**

Problems

License Plates

How many 7-character license plates are there that are all different letters and they are in alphabetical order?

$$\binom{26}{7}$$

How many 7-character license plates are there from all capital letters and they are in alphabetical order? They can be duplicates, so AAAAAAA is valid
Use stars and bars! x_A, x_B, x_C represents the number of appearances of each letter. They all must sum to 7
$n=7,\ k=26$

$$\binom{7+26-1}{7} = \binom{32}{7}$$

In general

$$\begin{aligned} n &= \text{string length} \\ k &= \text{size of alphabet} \end{aligned}$$

Rotational Symmetries

For an nth polygonal prism, there are $2n$ rotational symmetries

Basketball

There are 18 different basketball players. How many ways can they organize themselves into 3 games each with 3 players versus 3 players?

$$\frac{18!}{(3!3!3!3!3!3!)(2!2!2!)3!}$$

$\binom{18}{3}\binom{15}{3}$ double counts, so we need to divide by 2!. We need to divide by 3! too because each game is treated as distinct

Integer Equation Bounded

$$\begin{aligned} |U| &= |\text{all vars } \leq 7| + |\text{at least 1 var } \geq 8| \\ |\text{all vars } \leq 7| &= |U| - |\text{at least 1 var } \geq 8| \end{aligned}$$

$$|U| = \binom{35}{5}$$

|at least 1 var being at least 8| = |1 var at least 8|+|2 va
For each subset, "p var(s) at least 8", we must multiply the arrangements by $\binom{6}{p}$ since we need to choose where the "at least" variables are
For $p=1$

$$\begin{aligned} (a_1+8)+a_2+a_3+a_4+a_5+a_6 &= 30 \\ a_1+a_2+a_3+a_4+a_5+a_6 &= 22 \end{aligned}$$

$$\binom{6}{1}\binom{27}{5}$$

For $p=2$

$$\begin{aligned} (a_1+8)+(a_2+8)+a_3+a_4+a_5+a_6 &= 30 \\ a_1+a_2+a_3+a_4+a_5+a_6 &= 14 \end{aligned}$$

$$\binom{6}{2}\binom{19}{5}$$

For $p=3$

$$\begin{aligned} (a_1+8)+(a_2+8)+(a_3+8)+a_4+a_5+a_6 &= 30 \\ a_1+a_2+a_3+a_4+a_5+a_6 &= 6 \end{aligned}$$

$$\binom{6}{3}\binom{11}{5}$$

And for $p=4$ to $p=6$, there's 0 solutions since the RHS would be negative
But we need to be careful because we would overcount if we just add up all the sets. The solutions for $p=2$ are INCLUDED in $p=1$, and same for $p=3$, so we need to apply Principle of Inclusion Exclusion.
Recall that for PIE, if you're accounting for an ODD number of sets, then you subtract, but EVEN you add
so we add $p=1$ and $p=3$, but subtract $p=2$
Doing all of this, we get

$$\begin{aligned} &|\text{at least 1 var being at least 8}| = \\ &\binom{6}{1}\binom{27}{5} - \binom{6}{2}\binom{19}{5} + \binom{6}{3}\binom{11}{5} \end{aligned}$$

And plugging this back into our base equation we get

$$|\text{all vars } \leq 7| = \boxed{\binom{35}{5} - \binom{6}{1}\binom{27}{5} + \binom{6}{2}\binom{19}{5} - \binom{6}{3}\binom{11}{5}}$$